NON-TRIVIAL

A Multifaceted Index for Carbon Capture Site Evaluation – Cheng-You Ho

Executive Summary (~250 words)

- The problem you're addressing and why it matters
- Your research question and approach
- How your work could contribute to solving the problem
- Why you're well-positioned to do this research

Climate change is one of the most pressing issues today, with increasing concentrations of greenhouse gases as a primary cause. Carbon capture and storage (CCS) offers a promising solution by capturing CO_2 emissions and storing them underground. New storage sites are actively being searched. The selection of CCS sites depends on both geological properties, such as depth and porosity of the reservoir, and non-technical factors, including regulatory frameworks and socio-economic criteria. Current research, however, tends to focus on the geological aspects, and research regarding the latter is often disconnected from technical factors. Also, there isn't a comprehensive measure that combines both technical and non-technical effects.

To bridge this gap, I aim to design a multifaceted, quantitative index that reflects the overall effectiveness of CCS sites. By utilizing open databases for CCS projects, geological surveys, economic and regulatory indicators, etc., I will employ statistical methods such as multi-criteria decision analysis (MCDA) and Monte Carlo to evaluate a broad range of factors. In addition, I plan to analyze existing CCS projects and identify the key aspects that drive cost-effective CCS implementation. This integrated index could then help decision-makers take non-technical factors into account and locate the most suitable sites for future projects.

I have research experience in physics and a strong foundation in science and math. Moreover, I'm familiar with Python and data analysis, which makes me well-positioned for this project.

Problem analysis (~500 words)

Problem statement

- What specific problem are you addressing?
- What makes this a pressing problem?
- What aspects of the problem will you focus on and why?

Current landscape

- What are the main approaches to addressing this problem?

- What important gaps or limitations exist?
- What unique opportunities have you identified?
- What evidence supports your analysis?

Problem Statement

The issue addressed in this proposal is the lack of a comprehensive, quantitative index for evaluating carbon capture and storage (CCS) sites that integrate both technical and non-technical factors. While existing studies often focus on geological properties such as reservoir depth, porosity, permeability, and caprock integrity, they often neglect non-technical aspects, such as regulatory frameworks, economic conditions, and socio-economic factors like community acceptance.

These aspects are essential. For instance, if a carbon storage site is too close to a residential area, it may lead to extra safety-related costs or rejections from residents, leading to delays or even cancellations regardless of the site's excellent geological characteristics. Furthermore, a site located in a region with weak regulatory oversight could result in environmental risks, such as undetected CO_2 leakage, ultimately undermining the project's safety and effectiveness.

CCS site evaluation is a pressing problem since the technology is currently being actively developed, companies are searching for new sites for future projects, and climate change is becoming increasingly urgent. I will focus on both technical and non-technical factors because while there have been decades of research regarding the former, combining it with non-technical factors remains underexplored.

Current Landscape

Historically, CCS site evaluations have focused on geological aspects. Methods such as geologic screening, risk-based ranking, and reservoir modeling are used to identify suitable sites. Frameworks such as Oldenburg's risk-based scoring system prioritize minimizing risk when ranking sites. There have also been case studies on modeling potential CCS sites with seismic data. However, these technical evaluations are often isolated from non-technical analyses.

On the non-technical side, regulatory assessments, economic feasibility studies, and socio-economic evaluations had been carried out. For example, regulations of The U.S. Environmental Protection Agency (EPA) will affect whether a particular CCS site in the U.S. is suitable. Furthermore, socio-economic effects of carbon capture in China have been investigated. These data provide valuable insights into the feasibility of a CCS project beyond geological aspects. Yet, they are rarely combined with technical data to form a unified decision-making tool. This creates a gap. Although we understand both components individually, there lacks a standardized model to combine them into a quantitative index.

Non-technical aspects are often neglected compared to technical factors, and this has brought about losses in past projects. Petra Nova, once a flagship U.S. CCS project, was suspended in 2020 due to financial difficulties exacerbated by low oil prices. Although its storage site, the Frio Formation, was a feasible location, the project wasn't successful due to economic factors. Similarly, a CCS project led by the Pathways Alliance in Canada faced

financial challenges due to the large distance that carbon dioxide needed to travel before reaching the capture site.

Studies show that 80% of commercial CCS projects ended in failure. If non-technical factors could be taken into account with, for example, a scoring system, and combined with technical aspects via a quantitative index, CCS projects could be more effective and more likely to succeed.

Research plan (~750 words)

Research question(s)

- Primary research question
- Sub-questions

Methodology

- Research methods you'll use
- Why these methods are appropriate
- Data sources and analytical approaches

Primary research question:

How can a multifaceted, quantitative index be developed to evaluate CCS sites by integrating both technical and non-technical factors?

Sub-questions:

1. What technical factors (e.g., geological properties) are most critical for CCS site suitability?
2. What non-technical factors (e.g., regulatory, economic, socio-economic) are most critical for the feasibility and long-term viability of CCS projects?
3. How can non-technical factors be quantified?
4. How can these diverse factors be integrated into a framework for evaluating CCS sites?
5. What weight should be assigned to each factor? In other words, what is the relative significance of each factor?
6. How does the proposed index compare with historical CCS project successes and failures?

Methodology

Research Methods and Analytical Approaches

First, I will perform a literature review and identify the most critical factors for CCS site selection. The result obtained from the literature is only qualitative.

With data from current and past CCS projects, multiple-criteria decision analysis (MCDA) will be used to combine technical factors and my scoring for non-technical aspects. MCDA is a structured decision-making method used to evaluate and rank multiple alternatives (CCS sites) based on several criteria (technical and non-technical factors). I will follow the steps below:

1. **Extracting key factors and normalizing data**:

To perform a quantitative analysis, technical and non-technical factors from databases will be extracted. The technical factors, such as reservoir depth, porosity, permeability, etc., are already quantified. These data will be obtained from geological surveys. To make data analysis more convenient, I will normalize each factor to a value from 0 to 1. Note that the scaling may not be linear. For example, if a reservoir depth threshold is set such that any depth beyond it is considered equally favorable, I would choose a function such as hyperbolic tangent instead.

Non-technical aspects, on the other hand, do not have quantified data available. Therefore, a scoring system that assigns a score from 0 to 1 for each factor based on policy reports, socio-economic indicators, etc. will be developed, with a higher score corresponding to better feasibility of the carbon capture site.

2. **Assigning weights**

Weights will be assigned to each factor with MCDA. The weighting process will be structured in two levels. At the first level, weights will be assigned to two main categories: technical factors and non-technical factors. At the second level, the relative importance of individual subfactors within each category will be adjusted. The goal of assigning weights is to identify the primary factors that affect the cost of capturing 1 ton of CO_2. After the weights are determined, the weighted score S will be calculated as:

$$S = \sum_{i=1}^{n} w_i X_i$$

where w_i and X_i represent the weight and value of the i-th factor, respectively. This score is the quantitative index that I want to develop.

3. **Sensitivity analysis**

To quantify the uncertainties of the weights, I will use Monte Carlo simulations. Using Python, thousands of random samples will be generated, each with a different set of weights for each factor. For each set, the weighted score S for each site will be calculated, and sites will be ranked based on the score. Then, the sensitivity of each factor could be analyzed. For instance, if a factor's weight fluctuates significantly between samples but has little effect on the final ranking of sites, it suggests that the factor has a lower impact on CCS site selection. Conversely, if small changes in a factor's weight cause major shifts in site rankings, it indicates that the factor is highly sensitive and plays a crucial role in determining suitability.

4. **Identifying potential sites**

The quantitative index can be calculated on locations where geological data and non-technical data are available. Thus, by calculating the weighted score and comparing it to that of existing carbon capture sites, potential sites for future CCS projects could be identified.

Data Sources

1. **CCS Project Databases**

- NETL's Worldwide CCS Database (free): A comprehensive spreadsheet that includes the scale, cost, location, etc. of over 400 CCS projects worldwide.
- Global CCS Projects Database by Thunder Said Energy ($499): Tracks over 400 CCS projects. Data includes project sizes, capital expenditures, and capacities.

2. **Geological Databases**

- National Carbon Sequestration Database and Geographic Information System (NATCARB).
- Geologic Carbon Dioxide Sequestration Interactive Map.

3. **Non-technical Data**

- International Energy Agency (IEA) reports and Policies Database.
- Government regulations, such as those of The U.S. Environmental Protection Agency (EPA).
- Socio-economic studies, such as this report.
- Economic indicators from World Bank Open Data.
- Human Development Reports.

Theory of change (~500 words)

Impact pathway

- Direct outputs of your research
- How these outputs lead to changes
- Who needs to do what differently
- How your project will accomplish that
- Ultimate impact on the problem

Risk assessment

- Key assumptions in your theory of change
- Major uncertainties or challenges
- Potential unintended consequences
- Mitigation strategies

Impact pathway

Direct outputs and changes

The primary output of this research is a multifaceted, quantitative index that integrates both technical and non-technical factors for evaluating CCS sites. This approach addresses the high failure rate of CCS projects, with studies indicating that approximately 80% of such initiatives have not achieved their intended outcomes, often due to non-technical factors such as financial viability, regulatory challenges, and public opposition. By utilizing this integrated index, stakeholders can make data-driven and holistic decisions regarding CCS site selection, ensuring that projects are not only geologically sound but also economically and socially feasible.

The index will offer a standardized decision-making tool that allows policymakers, industry leaders, and investors to optimize costs and improve site selection efficiency. By incorporating both technical and non-technical assessments, it will help identify sites that have the best chance of long-term success.

Who needs to do what differently

Regulators and policymakers should integrate the index into policy frameworks and use its evaluations to assess and audit CCS project proposals before approval. This will ensure that CCS projects selected for development align with both technical feasibility and regulatory standards. Industry stakeholders should also adopt this index as a part of their site selection process.

How your project will accomplish that

After the index is developed, I will share the project through my presentation video and final research paper through Non-Trivial. If possible, I would publish my work in journals related to renewable energy and climate change, such as Renewable Energy and Carbon Capture Science & Technology.

Ultimate impact

The impact of this project is to help decision-makers select CCS sites, thus reducing the failure rate of CCS initiatives, and enhancing the efficacy of CCS as a climate mitigation strategy. Ultimately, this will help reduce the net emission of greenhouse gases and support global climate goals.

Risk assessment

I assumed that data are available for both technical and non-technical factors across potential CCS sites. While free data exist, they may be incomplete or outdated, and high-quality datasets could come at a significant cost. To mitigate this risk, I will ensure that most of the data is extracted from trustworthy sources such as NETL's Worldwide CCS Database and supplement gaps with peer-reviewed research and industry reports.

I also assumed that non-technical factors, such as economic conditions and community sentiments, remain stable over time, though they may fluctuate, potentially impacting the index's long-term effectiveness. To address this, if time permits, I will implement an update mechanism that periodically reassesses the weight of non-technical factors based on new data from new research and reports.

Furthermore, when this index is introduced to stakeholders, they may be resistant to adopting this novel methodology. Thus, if they have doubts about my work, I will present them with my evaluation results for past CCS projects. Additionally, I will calculate the index on existing case studies, and notify the original authors of my assessments to encourage collaboration and feedback.

Implementation plan (~250 words)

Timeline

- Major milestones & weekly targets
- Key dependencies
- Contingency planning

Resources

- Required data, tools, and access
- Skills and expertise needed
- Support and mentorship plans

Timeline Table

Week	Major Milestone	Dependencies	Contingency Plan
1	Literature Review & Factor Identification	Academic papers	Adjust scope if factors are difficult to identify
2	Data Collection & Normalization	Geological databases (NATCARB, USGS)	Adjust scope if data is unavailable
3	Scoring Non-Technical Factors	Policy data, socio-economic indicators	Use rankings from past research if data is limited
4	Assigning Weights with MCDA	Historical CCS project databases, cost data	-
5	Computing Index on Existing Projects	Weight assignment from MCDA	-
6	Sensitivity Analysis with Monte Carlo	Python programs	-
7	Validating Index Effectiveness & Preparing Report	Historical CCS project databases	Rely on qualitative comparisons if needed
8	Finalizing, Writing Report & Filming Video	-	-

Resources

The required data is already listed above. The skills and expertise needed include background knowledge in math, and statistical methods such as MCDA and Monte Carlo. Data analysis and running Monte Carlo simulations will require knowledge of Python. Feedback from industry experts and academic professionals would also be beneficial to this project.

Personal fit (~250 words)

Capabilities

- Relevant skills and experience

- Access to required resources
- Understanding of the problem space

Commitment

- Why you care about this problem
- How this fits your longer-term goals

Capabilities

I have written two research papers on my work in physics and astrophysics, both of which involved Python-based data analysis and visualization to present findings. The physics paper required simulating analytical equations, while my astrophysics project involved working with the Gaia database, extracting astronomical data, and applying regression techniques to identify patterns. These experiences provide me with the technical skills needed for this project.

Beyond research, I have demonstrated strong proficiency in science and math by winning a silver medal at the International Junior Science Olympiad and reaching the finals of the national Physics Olympiad. These achievements reflect my problem-solving skills, analytical thinking, and the ability to apply scientific principles effectively.

Commitment

During my first year of high school, I made a video on carbon capture for a renewable energy competition. I became so fascinated by this technology that I spent hours creating an animation to demonstrate its working principles. This initial exposure led me to attend multiple academic talks by professors discussing recent advancements in CCS. These experiences have inspired me to contribute to this topic at Non-Trivial.

This project also aligns with my long-term goal of becoming a physicist or engineer, where I will apply analytical approaches to solve real-world problems. CCS involves using physics to model the technical factors, finding the optimal solution with statistics, and improving environmental sustainability. I'm excited to embark on this project and explore the intersection of these fields.

Appendix

Alternative approaches considered

- Other project ideas you explored
- Why you chose this approach instead
- Key trade-offs you're making

Allocating Carbon Credits in Blockchain-Enabled Emissions Trading Systems

Carbon credits are traded on Emissions Trading Systems (ETS). Since blockchain is decentralized, transparent, and secure, it can enhance ETS. These credits are allocated to

companies via certain frameworks. However, existing allocation methods such as grandfathering, benchmarking, and auctioning could have inefficiencies when applied to decentralized systems. Thus, I hope to review traditional allocation methods and design a hybrid mechanism (e.g. implementing multiple methods at the same time but with different weights) that ensures fairness, efficiency, and transparency when trading carbon credits.

I'm passionate about applying blockchains to carbon credits trading. However, simulating an allocation strategy may require advanced knowledge of smart contracts, auction mechanisms, and how blockchains work in detail. Therefore, although this approach is feasible, I think it is more practical to stick to my current approach to CCS.

Risk Modeling for Geological Storage

Capturing carbon dioxide within materials bears the risk of the gas being released. It is important to analyze what types of material and environmental conditions (temperature, humidity, etc.) are best for carbon capture. Also, it is critical to estimate the impact on the world if a storage site releases a significant amount of carbon dioxide by accident.

While a lot of research has been done regarding local environmental impacts, there is a need for assessing the risk and impact of potential large-scale CO_2 releases from storage sites. However, considering that this issue is less neglected, I chose my current approach instead.

References

In addition to the links embedded in the proposal document:

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- Parhamfar, M., Sadeghkhani, I., & Adeli, A. M. (2024). Towards the net zero carbon future: A review of blockchain-enabled peer-to-peer carbon trading. *Energy Science & Engineering, 12*(3), 1242–1264. https://doi.org/10.1002/ese3.1697
- Mukherjee, A., & Chatterjee, S. (2023). *Carbon capture, utilization and storage (CCUS): Technology gaps and international collaboration*. Dastur Energy & NTPC Ltd. https://powermin.gov.in/sites/default/files/uploads/CCUS_Technology_Gaps_and_International_Collaboration_G20_Circulation.pdf